Homework 11, Week 14 2023.12.06 Due 2023.12.13 L04.04, L04.07, L04.13, L04.18, L04.20

★ L04.04

已知质点在有心力作用下的轨道是通过力心的圆，试证质点所受有心力是 ρ^{-5} 的吸引力

Solution

两个基本方程：$\begin{cases} m r^2 \dot{\phi} = \ell \\ \dot{r} = \beta \left[\frac{2}{m}(E - U_{\text{eff}})\right]^{1/2} \end{cases}$，　$U_{\text{eff}} = U(r) + \frac{\ell^2}{2 m r^2}$

两式相比：$\frac{d r}{d \phi} = \beta \frac{m r^2}{\ell} \sqrt{\frac{2}{m}(E - U_{\text{eff}})}$　　接着变量代换：$r = \frac{1}{u}$

$$\frac{d u}{d \phi} = -\frac{\beta}{\ell} \sqrt{2 m \left[E - U_{\text{eff}}\left(\frac{1}{u}\right)\right]} \quad \Longrightarrow \quad U(r) = E - \frac{1}{2 m}\left(\frac{d u}{d \phi}\right)^2 \ell^2 - \frac{l^2}{2 m} u^2$$

轨道：$r = 2 R \cos \phi$,　$u = \frac{1}{2 R \cos \phi}$ 代入得

$$U = -\frac{\ell^2}{8 m R^2 \cos^4 \phi} = -\frac{\ell^2}{8 m R^2}(2 R u)^4 = -\frac{2 R^2 \ell^2}{m r^4} \quad \Longrightarrow \quad F(r) = -\frac{d U}{d r} = -\frac{8 R^2 \ell^2}{m r^5}$$

```
u = 1/(2 R Cos[ϕ]);
U = -1/(2 m) D[u, ϕ]^2 l0^2 - l0^2/(2 m) u^2;
Simplify[U]
```

$$-\frac{l0^2 \operatorname{Sec}[\phi]^4}{8 m R^2}$$

★ L04.07

质点在有心力常中运动，场中势能为 $V(\rho) = -k \frac{e^{-\alpha \rho}}{\rho}$，其中 k 和 α 都是正的常数，试求力的表达式 $F(\rho)$，并对质点运动情况作定性讨论。最后求质点作圆运动的周期和径向小振动的周期。

Solution

$$F = -\frac{d V}{d \rho} = -\frac{k(1 + \alpha \rho) e^{-\alpha \rho}}{\rho^2}$$

有效势场：$U_{\text{eff}} = V(\rho) + \frac{\ell^2}{2 \mu \rho^2} = -\frac{k e^{-\alpha \rho}}{\rho} + \frac{\ell^2}{2 \mu \rho^2}$ 中的一维运动

圆周运动：$\left.\frac{\partial U_{\text{eff}}}{\partial \rho}\right|_{\rho = R} = 0 \quad \Longrightarrow \quad \frac{k(1 + \alpha R) e^{-\alpha R}}{R^2} - \frac{\ell^2}{\mu R^3} = 0$　　(1)

稳定性：$\left.\frac{\partial^2 U_{\text{eff}}}{\partial \rho^2}\right|_{\rho = R} > 0 \quad \Longrightarrow \quad \frac{3 \ell^2}{\mu R^4} - \frac{k\left(2 + 2 \alpha R + \alpha^2 R^2\right) e^{-\alpha R}}{R^3} > 0$　　(2)

联立式 (1) 和 (2) 可得：$\begin{cases} 1+\alpha R-\alpha^2 R^2>0 \\ R<\dfrac{1+\sqrt{5}}{2\alpha} \end{cases}$

圆轨道附近：$U_{\text{eff}}=U(R)+\dfrac{1}{2}(\rho-R)^2\left.\dfrac{\partial^2 U_{\text{eff}}}{\partial\rho^2}\right|_{\rho=R}+\ldots$

相当于在一维简谐势场 $\dfrac{1}{2}(\rho-R)^2\left.\dfrac{\partial^2 U_{\text{eff}}}{\partial\rho^2}\right|_{\rho=R}$ 中的运动，小振动角频率为 $\omega=\sqrt{\kappa/\mu}$

$\kappa=\left.\dfrac{\partial^2 U_{\text{eff}}}{\partial\rho^2}\right|_{\rho=R}$， 小振动周期 $\tau=\dfrac{2\pi}{\omega}=2\pi\sqrt{\dfrac{\mu R^3 e^{\alpha R}}{k\left(1+\alpha R-\alpha^2 R^2\right)}}$

圆周运动周期：

单位时间扫过的面积：$dA=\dfrac{1}{2}\rho^2\dot{\phi}\,dt=\dfrac{\ell}{2\mu}dt$ $\qquad\dfrac{dA}{dt}=\dfrac{\ell}{2\mu}$

圆面积：$A=\pi R^2$ $\quad\Longrightarrow\quad T=\dfrac{A}{dA/dt}=\dfrac{2\mu\pi R^2}{\ell}$ 由 (1) 可将 ℓ 表示为 R 的函数

故：$T=2\pi\sqrt{\dfrac{\mu R^3 e^{\alpha R}}{k(1+\alpha R)}}$

★ L04.13

质点在有心力场中运动，这有心力场基本上是平方反比引力，但附加有 ρ^{-3} 的摄动

$F(\rho)=-\dfrac{k^2}{\rho^2}+\dfrac{C}{\rho^3}$ 这里 k^2 和 C 是常数

试证质点的轨道方程可表示为：$\rho=\dfrac{a\left(1-\varepsilon^2\right)}{1+\varepsilon\cos\alpha\varphi}$.

这是椭圆，但因 $\alpha\neq 1$，这椭圆是进动的。试在 $\alpha\approx 1$ 条件下推导椭圆进动速率公式。

Solution

由力求轨道，可以用 Binet 方程，也可从头计算。

有心势场：$V(\rho)=-\dfrac{k^2}{\rho}+\dfrac{C}{2\rho^2}$ 两基本方程：$\begin{cases} m\rho^2\dot{\phi}=\ell \\ \dot{\rho}=\beta\left[\dfrac{2}{m}(E-U_{\text{eff}})\right]^{1/2} \end{cases}$

有效势场：$U_{\text{eff}}=V(\rho)+\dfrac{\ell^2}{2\mu\rho^2}=-\dfrac{k^2}{\rho}+\dfrac{C}{2\rho^2}+\dfrac{\ell^2}{2\mu\rho^2}$

轨道方程：$\dfrac{d\varphi}{d\rho}=\dfrac{\ell}{m\rho^2}\dfrac{1}{\beta\left[\dfrac{2}{m}(E-U_{\text{eff}})\right]^{1/2}}$

$$\Longrightarrow\quad \varphi=\beta\int\frac{\ell}{m\rho^2}\frac{d\rho}{\left[\dfrac{2}{m}(E-U_{\text{eff}})\right]^{1/2}}=\beta\int\frac{1}{\rho^2}\frac{m\,h\,d\rho}{\left[2m\left(E+\dfrac{k^2}{\rho}-\dfrac{C}{2\rho^2}-\dfrac{\ell^2}{2m\rho^2}\right)\right]^{1/2}}$$

令 $\begin{cases} \ell=m\,h \\ \rho=1/u \end{cases}$

$$\Longrightarrow\ \varphi=-\beta\int\frac{m\,h\,du}{\sqrt{2\,m}\left[E+k^2u-\left(\frac{C}{2}+\frac{m\,h^2}{2}\right)u^2\right]^{1/2}}$$

$$=-\beta\int\frac{du}{\left[\frac{2\,E+2\,k^2\,u}{m\,h^2}-\left(1+\frac{C}{m\,h^2}\right)u^2\right]^{1/2}}=-\beta\int\frac{du}{\alpha\left[\frac{2\,E}{m\,h^2\,\alpha^2}+\frac{2\,k^2\,u}{m\,h^2\,\alpha^2}-u^2\right]^{1/2}}$$

$$=-\beta\int\frac{du}{\alpha\left[\left(\frac{\varepsilon}{p}\right)^2-\left(u-\frac{1}{p}\right)^2\right]^{1/2}}$$

其中：$\alpha=\sqrt{1+C/m\,h^2}$，$p=m\,\alpha^2\,h^2/k^2$，$\varepsilon=\sqrt{1+2\,p\,E/k^2}$

取 $\rho_{\min}$ 处 $\varphi=0$，随 φ 增加，ρ 增大，$u=\frac{1}{\rho}$ 减小，故应取 $\beta=1$

积分得：$\varphi=\frac{1}{\alpha}\arccos\frac{(1/\rho-1/p)}{\varepsilon/p}\quad\Longrightarrow\quad\rho=\frac{p}{1+\varepsilon\cos\alpha\,\varphi}=\frac{a(1-\varepsilon^2)}{1+\varepsilon\cos\alpha\,\varphi}$，　其中 $a=\frac{p}{1-\varepsilon^2}$

考虑束缚态，$E<0$,

如果 $C=0$，则 $\alpha=\sqrt{1+C/m\,h^2}=1$，

$\rho=\frac{a(1-\varepsilon^2)}{1+\varepsilon\cos\varphi}$ 径向完成一个周期，φ 变化 $2\,\pi$，构成封闭轨道

如果 $C\neq0$，则 $\alpha=\sqrt{1+C/m\,h^2}\neq1$

$\rho=\frac{a(1-\varepsilon^2)}{1+\varepsilon\cos\alpha\,\varphi}$ 径向完成一个周期，$\alpha\,\varphi$ 变化 $2\,\pi$，即 $\rho_{\min}$ 转过了 $2\,\pi\left(\frac{1}{\alpha}-1\right)$

相当于椭圆取向在旋转。

一个径向周期（从 $\rho_{\min}$ 到 $\rho_{\max}$ 在回到 $\rho_{\min}$）椭圆转过了 $\Delta\theta=2\,\pi\left(\frac{1}{\alpha}-1\right)$

思考：为何不是 $\Delta\theta=2\,\pi\left(1-\frac{1}{\alpha}\right)$

接着需要求径向周期 T

从 $\rho_{\min}$ 到 $\rho_{\max}$ 在回到 $\rho_{\min}$，扫过的面积 $A=\int_0^{2\,\pi/\alpha}\frac{1}{2}\,\rho^2\,d\varphi$，利用 $\rho=\frac{p}{1+\varepsilon\cos\alpha\,\varphi}$

$$A=\frac{1}{2}\int_0^{2\,\pi/\alpha}\frac{p^2\,d\varphi}{(1+\varepsilon\cos\alpha\,\varphi)^2}=\frac{1}{2\,\alpha}\int_0^{2\,\pi}\frac{p^2\,d\phi}{(1+\varepsilon\cos\phi)^2}=\frac{1}{\alpha}A_1$$

A_1 为椭圆 $\frac{p\,d\phi}{(1+\varepsilon\cos\phi)}$ 的面积，$A_1=\frac{1}{2}\int_0^{2\,\pi}\frac{p^2\,d\phi}{(1+\varepsilon\cos\phi)^2}=\frac{\pi\,p^2}{(1-\varepsilon^2)^{3/2}}$，

```
Integrate[1/(2 (1 + ϵ Cos[φ])^2), {φ, 0, 2 π}, Assumptions → {0 < ϵ < 1}]
```

$$\frac{\pi}{(1-\epsilon^2)^{3/2}}$$

单位时间扫过的面积 $dA = \frac{1}{2}\rho^2 \dot{\phi}\, dt \implies \frac{dA}{dt} = \frac{\ell}{2m} = \frac{h}{2}$

故径向周期 $T = \frac{A_1/\alpha}{dA/dt} = \frac{\pi p^2}{\alpha (1-\varepsilon^2)^{3/2} h/2} = \frac{2\pi a^2 \sqrt{1-\varepsilon^2}}{\alpha h}$

进动角速度：$\omega = \frac{\Delta\theta}{T} = \frac{(1-\alpha) h}{a^2 \sqrt{1-\varepsilon^2}}$

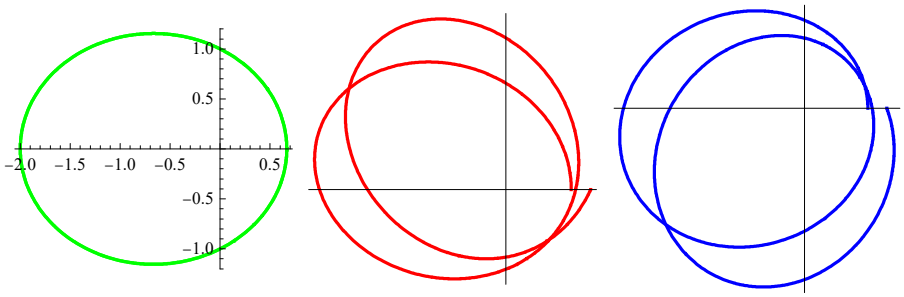
$$1-\alpha = 1 - \sqrt{1 + C/m h^2} \approx -\frac{C}{2 m h^2} \implies \omega = -\frac{C}{2 m h} \frac{h}{a^2 \sqrt{1-\varepsilon^2}}$$

负号表示 当 $C > 0$（即 $\alpha > 1$）时椭圆进动发现与质点绕力心转动发现相反。（参见以下动态图）

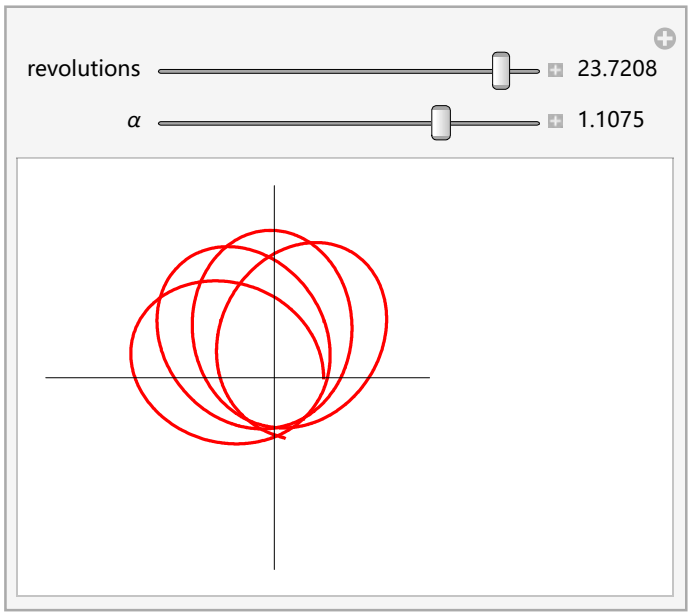
Mathematica code 显示 α < 1, α = 1, α > 1 参数下的轨迹

```
Quit
```

```
rho = p/(1 + ε Cos[α φ]);
num1 = {p → 1, ε → 0.5, α → 1};
num2 = {p → 1, ε → 0.5, α → 1.1};
num3 = {p → 1, ε → 0.5, α → 0.9};
g1 = PolarPlot[rho /. num1, {φ, 0, 4 π}, PlotStyle → {Green}, ImageSize → {150, 150}];
g2 = PolarPlot[rho /. num2, {φ, 0, 4 π},
   PlotStyle → {Red}, Ticks → None, ImageSize → {150, 150}];
g3 = PolarPlot[rho /. num3, {φ, 0, 4 π}, PlotStyle → {Blue},
   Ticks → None, ImageSize → {150, 150}];
Grid[{{g1, g2, g3}}]
```



```
p = 1;  ε = 0.5;
Manipulate[Module[{rho},
  rho[φ_, α_] := 1/(1 + ε Cos[α φ]);
  PolarPlot[rho[φ, α], {φ, 0, φmax}, PlotStyle → {If[α > 1, Red, If[α == 1, Green, Blue]]},
   Ticks → None, PlotRange → {{-3, 2}, {-2.5, 2.5}}, ImageSize → {200, 200}]],
  {{φmax, 0.1, "revolutions"}, 0.1, 8 π, Appearance → "Labeled"},
  {{α, 1, "α"}, 0.8, 1.2, Appearance → "Labeled"},
 ControlPlacement → Top, SaveDefinitions → True]
```



★ L04.18 需用到周一（12.11）上课的内容

试研究立方反比斥力 $F(\rho) = k\,\rho^{-3}$ 的散射问题。试证明散射截面 $\sigma(\theta) = \dfrac{k\,\pi^2}{2\,E}\,\dfrac{\pi-\theta}{\theta^2(2\,\pi-\theta)^2\,\sin\theta}$

Solution

两基本方程：$\begin{cases} m\,\rho^2\,\dot{\phi} = \ell \\ \dot{\rho} = \beta\left[\dfrac{2}{m}\,(E - U_{\text{eff}})\right]^{1/2} \end{cases}$ ， $U_{\text{eff}}(r) = U(r) + \dfrac{\ell^2}{2\,m\,\rho^2}$， $U(r) = \dfrac{k}{2\,\rho^2}$

无穷远入射 $E = \dfrac{1}{2}\,m\,v^2$， $\ell = m\,v\,b \implies E = \dfrac{\ell^2}{2\,m\,b^2}$

$$\implies \frac{d\phi}{d\rho} = \frac{\beta\,\ell}{m\,\rho^2}\,\frac{1}{[2\,m\,(E - U_{\text{eff}})]^{1/2}} \qquad \implies \phi = \sqrt{\frac{\ell^2}{2\,m}}\int\frac{\beta\,d\rho}{\rho^2[E - U_{\text{eff}}]^{1/2}}$$

在 $\rho = \rho_{\min}$ 处，$\dot{\rho} = 0$，能量 E 等于有效势能： $E = U(\rho_{\min}) + \dfrac{\ell^2}{2\,m\,\rho_{\min}^2} = \dfrac{k}{2\,\rho_{\min}^2} + \dfrac{b^2\,E}{\rho_{\min}^2}$

可求得：$\dfrac{1}{\rho_{\min}} = \dfrac{\sqrt{2}\,E}{(k + 2\,b^2\,E)^{1/2}}$

$$\varphi = \phi = \sqrt{\frac{\ell^2}{2\,m}} \int_{\rho_{\min}}^{\infty} \frac{d\rho}{\rho^2[E - U_{\text{eff}}]^{1/2}} = b\sqrt{E} \int_{\rho_{\min}}^{\infty} \frac{d\rho}{\rho^2[E - U_{\text{eff}}]^{1/2}} \quad \text{利用了 } \ell^2 = 2\,m\,b^2\,E$$

$$= \int_{\rho_{\min}}^{\infty} \frac{b\,d\rho}{\rho^2\sqrt{1 - (k/E + 2\,b^2)/(2\,\rho^2)}} \quad \text{其中 } U_{\text{eff}} = \frac{k}{2\,\rho^2} + \frac{\ell^2}{2\,m\,\rho^2} = \frac{k}{2\,\rho^2} + \frac{b^2\,E}{\rho^2}$$

$$= \int_{\rho_{\min}}^{\infty} \frac{b\,d\rho}{\rho^2\sqrt{1 - \rho_{\min}^2/\rho^2}} = \frac{b\,\pi}{2\,\rho_{\min}}$$

```
Integrate[1/ρ^2/Sqrt[1 - ρmin^2/ρ^2], {ρ, ρmin, ∞}, Assumptions → {ρmin > 0}]
```

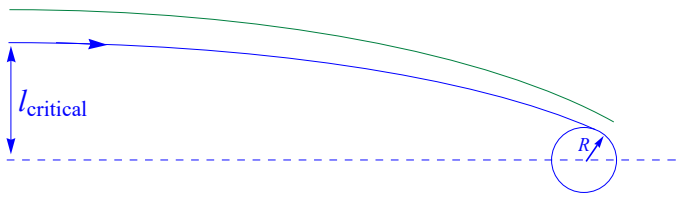
$$\frac{\pi}{2\,\rho\text{min}}$$

$$\theta = \pi - 2\,\varphi = \pi - \frac{b\,\pi}{\rho_{\min}} = \pi\left[1 - \frac{b}{(b^2 + k/2\,E)^{1/2}}\right] \implies b^2 = \frac{(\pi - \theta)^2}{\pi^2 - (\pi - \theta)^2}\,\frac{k}{2\,E}$$

所以，微分散射截面为 $\sigma(\theta) = \dfrac{1}{2\sin\theta}\left|\dfrac{d\,b^2(\theta)}{d\theta}\right| = \dfrac{k\,\pi^2}{2\,E}\,\dfrac{(\pi - \theta)}{\theta^2(2\,\pi - \theta)^2\sin\theta}$

★ L04.20 可能用到周一（12.11）上课的内容

俘获截面： 宇宙飞船关闭发动机，以速度 v_0 航行。飞船的目标是远处某一行星。试求临界瞄准距离 l_{critical}。只要瞄准距离小于这临界值 l_{critical}，就能在该行星上降落。$\pi\,l_{\text{critical}}^2$ 称为该行星的俘获截面。



Solution

无穷远入射 $E = \dfrac{1}{2}\,m\,v^2$，$\ell = m\,v\,b$

有效势 $U_{\text{eff}} = U(r) + \dfrac{\ell^2}{2\,m\,r^2} = -\dfrac{G\,M\,m}{r} + \dfrac{(m\,v\,b)^2}{2\,m\,r^2}$

被俘获，$E \geq U_{\text{eff}}(R) \implies \dfrac{1}{2}\,m\,v^2 \geq -\dfrac{G\,M\,m}{R} + \dfrac{(m\,v\,b)^2}{2\,m\,R^2}$

$$b^2 \leq \frac{2\,R^2}{m\,v^2}\left(\frac{1}{2}\,m\,v^2 + \frac{G\,M\,m}{R}\right), \qquad b \leq \left(R^2 + \frac{2\,R\,G\,M}{v^2}\right)^{1/2} = l_{\text{critical}}$$